Quick In-Class Challenge: Customer Purchase Analysis

Time: 10–15 minutes
Topic: Iterating Through Data Structures & Data Wrangling
Dataset: UCI_Online_Retail_7500row_practice.csv

Scenario

You are analyzing customer purchasing activity. Your goal is to identify customers who have made the highest-value purchases.

Challenge

1. Create a Customer Sales Dictionary

Create a dictionary where:

- **Key** = CustomerID
- **Value** = total SalesValue for that customer

For example:

```
{
  12345: 1250.50,
  67890: 845.25,
  ...
}
```

Hint:
Start with an empty dictionary:

```
customer_sales = {}
```

Loop through the rows and check whether the customer already exists in the dictionary.

2. Find the Highest-Spending Customer

Use your dictionary to find:

- The CustomerID with the highest total sales
- The corresponding total sales amount

Hint:

```
max(customer_sales, key=customer_sales.get)
```

3. Create a List of High-Value Customers

Create a list containing the CustomerID of every customer whose total sales are **greater than 1,000**.

Hint:

Use a loop:

```
high_value_customers = []


for customer, sales in customer_sales.items():
    # check the sales amount
```

⭐ **Bonus**

How many high-value customers did you find?

Can you display their CustomerIDs and total sales?

Restrictions

For this challenge, Focus on practicing **loops, dictionaries, lists, and DataFrames**.